

2(a) Area under F-x graph is w.d. in stretching bow. More potential energy is stored B1

Arrows gain more KE, has less deviation from intended flight path B1

OR

Arrows gain more KE, has farther range

2(b)(i) By counting squares under F-x graph, M1
Elastic potential energy = area under F-x graph
= 1.25 J × 64.5 (±2 squares)
= 81 J (78.1 J to 83.1J) A1

2(b)(ii) By Conservation of Energy, M1
Loss in elastic PE = Gain in KE for arrow
$$81 = \frac{1}{2} (3.5 \times 10^{-2}) v^2$$

 $v = 68.0 \text{ m s}^{-1}$ (66.8 J to 68.9 J) A1

3(a) B1